Section 0: Title of Project

Green Light for an Indie Studio

Section 1: Abstract/Introduction

Small game studios must make important decisions about game genre, pricing and scope before knowing how players will respond. This project proposes to investigate whether publicly available game data can provide useful evidence to support these decisions.

We will combine Steam game data with HowLongToBeat playtime data to examine the relationship between critic scores, player reception and player engagement. The study will investigate how well critic scores reflect player reception, whether controllable game characteristics can help explain player reception and completion, and how the amount of review evidence affects the reliability of player-reception estimates. We will also develop a completion ratio by comparing median Steam playtime with the typical time required to complete a game's main story. Statistical analysis will be conducted using Python, with the findings intended to provide practical insights for small game studios when making pre-release decisions.

Section 2: Overview

Problem Statement

A small studio interested in developing a game commits to a genre, a scope and a price range long before its release. However, it might have no way of knowing if players will like it until then.

Motivation

With less capital, smaller game studios have less flexibility with poor selling games as larger studios do. As such, the value of understanding future performance is greater.

They may have limited budgets to run experiments or buy market research, so secondary data is the main evidence available. Public evidence that a studio could learn from such as Steam / Metacritic reviews, however, do not close that gap. These scores exist only when somebody chooses to review (selection bias), and are often percentages without absolute counts stated (sampling variability).

Objectives

The objective of this project is to identify if a studio can make an informed decision about its key decisions on an upcoming game, in order to maximise consumer acceptance and, by extension, profit. By applying statistical modeling to Steam and HowLongToBeat data, we aim to identify key factors allowing the studio to make data-driven, pre-release decisions that minimize financial risk and maximize revenue.

Analytical Questions

1. What is a good metric for evaluating a game's success? We evaluate multiple possible metrics such as reviews, total playtime, completion rate.
2. Can we predict how successful a game will be? We study if, using the available data, we can build a model that holds some predictive power in predicting success.
3. What factors seem to be the most important for a game? We want to study if there are certain factors that are especially important that determine the success of a game, and highlight ones that a studio can control.
4. How do we translate these actionable insights into decisions that a business can make? For example, are there markets that are more important to user scores?
5. How does the number of player reviews affect the reliability of estimated player reception?

Context & Novelty

Some prior academic studies have researched estimating video game success. These s primarily relied on projecting sales figures, which can be a polarizing metric due to the difference in monetary expectations between different developers. Trněný [1] used an alternative indicator by measuring the volume of active players or owners post release. Bailey and Miyata [2] used Steam data to discover correlations between completion rates, user ratings, and game genres.

Similarly it is a standard to use median playtime as a metric for evaluating player engagement. Based on benchmark surveys of over 130 games by industry analyst Chris Zukowski " What is a good median play time for a demo?" [3], the results demonstrated that games in the highest lifetime revenue tiers exhibited higher playtimes during their pre-release demo phase (65 minutes median playtime). By relying on such engagement data, developers can identify what drives player satisfaction and make informed decisions on the project scope. Another instance of this was observed in P. Kępka and A. Strzelecki [4], where datasets consisting of online player surveys were used to predict player motivation. The results of this analysis demonstrated that a critical factor of player motivation included the openness of the game world. In the case for linear games, achievements were a factor in determining player motivation.

We propose the completion ratio as an alternative metric. Steam records how long the median owner played a game, while HowLongToBeat records how long that game typically takes to finish. A ratio of

1.00 means the median player spent the typical number of hours a game takes to complete, and above 1.00 means they went past it. Both datasets are well used on Kaggle. Partial pooling keeps low-visibility games in as weak evidence instead of filtering them out, and selection is treated as part of the question, including the match rate of the join itself.

As mentioned earlier, small studios have a lot less flexibility with games that face poor reception and sales post release compared to larger AAA studios. In such cases, it is critical to understand future performance ahead of release. By identifying quantifiable success metrics and evaluating factors that are within a studio's control, the proposed model will provide a relevant framework to assist developers in making informed pre-release decisions regarding the scope and genre.

Section 3: Data Sources

Dataset and Variables

The project will utilise two different datasets, both of which were sourced from Kaggle.

The first dataset consists of approximately 125,855 games published on the PC gaming platform Steam. The data was scraped using Steam Games Scrapper, an open source scrapper that extracts information from the Steam Web API.This dataset excluded additional assets published on Steam including: Downloadable content, episodes, music, and videos.

The second dataset consists of normalized and cleaned playtime statistics for video games collected from the website HowLongToBeat using a custom python web scraper. This dataset includes games across all platforms, DLCs and multiplayer-focused titles and it breaks down the user-reported playtime data for different playstyles:

- Main Story - Completing only the main objectives of the game
- Main + Side - Completing main objectives + some of the side content
- Completionist - 100% completion of the game
- All Styles - Average playtime across the three playstyles above
- Single-player/Co-op/Versus - Gamemode specific playtime

As the two datasets do not share a common identifier, they will be joined using normalised game titles and release year. The match rate will also be recorded to assess how many games can be successfully matched between the two datasets.

Data Dictionary

Source	Field	Type	Description
Steam	AppID, name, release_date	Id, Text, Date	Identity and game age
Steam	price	Numeric USD	Price band, and a covariate throughout
Steam	Positive, negative	Counts	Player reception and the sample size
Steam	recommendations	Counts	Store recommendations
Steam	metacritic_score	Numeric 0 to 100	Critic reception
Steam	Median_playtime_forever,	Minutes	Engagement, a SteamSpy estimate,

	average_playtime_forever		numerator of the completion ratio
Steam	Median_playtime_2weeks, average_playtime_2weeks	Minutes	Retention signal
Steam	estimated_owners	Bucketed Range	Audience size, a SteamSpy estimate
Steam	peak_ccu	Count	Concurrent audience
Steam	Genres, categories, tags	Lists	Genre and Mode
Steam	Developers, Publishers	Lists	Grouping for clustered errors and hierarchical models
Steam	achievements, dlc_count, supported_languages	Counts, List	Production scale proxies
Steam	Windows, Mac, Linux	Bool	Platform support
Steam	required_age	Numeric	Audience age band
HLTB	Id, name, release_date, release_year	Id, Text, Date	Join keys
HLTB	type	Categorical	Filter to game, excluding DLC and expansions
HLTB	Main_story, main_story_polled	Hours, Count	Denominator of the completion ratio, with its sample size
HLTB	main_plus_sides, completionist and poll counts	Hours, Count	Scope bands and the completionist gap
HLTB	Genres, Platform	Lists	Cross-check against Steam genres

Section 4: Methodology

Approach & Methodology

AQ1 - we will evaluate the benefits and limitations of different metrics in the context of a game studio. We evaluate the distribution of these metrics and their suitability for modelling.

AQ2 - We prepare the data by joining and cleaning them, processing them into a model-ready format (e.g. one hot encoding categorical variables) and split them into train and test datasets. We primarily evaluate a linear regression model for explainability, and explore if a frequentist or bayesian approach differs.

AQ3 - We evaluate the coefficients to identify the top performing variables, and construct a confidence interval on the coefficients. We build a smaller model using the top performing factors to see if a simpler model can be constructed.

AQ4 - Based on the inference of the results, we evaluate these in combination with factors that the studio can control. For example, we look at the language support to determine if, by extension, supporting more languages could lead to better reviews. However, there are other factors that are out of the studio's control that we must also evaluate, if they are able to even consider them.

AQ5 - We calculate the number of positive and negative reviews for each game and examine how the uncertainty around the estimated positive-review rate changes with the number of reviews. Games with smaller review counts are expected to provide less precise estimates of player reception than games with larger review counts.

Section 5: References

[1] “Machine learning for predicting success of video games,” Masaryk University Faculty of Informatics, 2017. [Online]. Available: https://is.muni.cz/th/k2c5b/diploma_thesis_trneny.pdf

[2] E. Bailey and K. Miyata, “Improving video game project scope decisions with data: An analysis of achievements and game completion rates,” *Entertainment Computing*, vol. 31, p. 100299, Mar. 2019, doi: 10.1016/j.entcom.2019.100299.

[3] “What is a good median play time for a demo? [Benchmark] – How To Market A Game,” Oct. 26, 2022. https://howtomarketagame.com/2022/10/26/what-is-a-good-median-play-time-for-a-demo-benchmark/

[4] P. Kępka and A. Strzelecki, “Exploring the factors influencing motivation and satisfaction of video game players,” *Journal of Electronic Gaming and Esports*, vol. 2, no. 1, Jan. 2024, doi: 10.1123/jege.2023-0026.